

Econometrics STATA Lab: Do-Files, Descriptive Statistics, and Joins

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Week 2

Acknowledgement: These slides draw on materials prepared by Kitak Kim and past teaching assistants.

Recap Week 1: Log file

- Make sure you are connected to your Google Drive.
- Set the directory to your Google Drive location **use_google**
- Or use a global macro **global wd "D:\PhotonUser\My Files\Google Drive\My Drive\AppData\Local\Temp\Stata"**.
- Create a new folder for week 2 **mkdir Week02**.
- Make this your directory **cd "\$wd\Week02"**
- Create a record of what you do in class **log using "wk2_log.log"**
- To close the log type **log close**
- This week we will see another way of 'recording' our code.

Do File

- Do files are scripts that store a sequence of STATA commands that will be executed in order.
- **Why do we need Do files?**
 - ① Record/save the exact commands that we ran for an analysis. This keeps work structured and we can share our work with collaborators.
 - ② Writing long lines of code in the command window is just tedious.
 - ③ Also helps us store the output of our commands in a structured manner (log file).
- In itself, a Do-File is a text file saved with a **.do** extension. It can be edited using any text editor.
- STATA does **not** run the commands written in the Do file immediately.
- Opening a Do file: Windows → Do-file Editor → New do-file editor.

Do File

Let's make a do file. Type the following into your code environment:

```
cd "$wd\Week02"  
bcuse affairs  
describe  
correlate naffairs ratemarr
```

Commenting your code

- Commenting your code helps keep your code neat and tidy and helps others understand the reason for writing a particular command.
- `*` in front of a line comments that complete line.
- If you want to comment out a chunk of code then use the `/* */`.
- Make sure to use commenting in your Problem Set and write your name and section using the commenting option.

Do File

This looks much nicer.

```
* My first do-file by Kitak
*make folder "EC2228\week2"
cd "$wd\Week02"
*load Affairs data
bcuse affairs
*obtain description of variables
describe
*calculate the correlation between happiness of
*marriage and number of affairs
correlate naffairs ratemarr
```

Log File

- There are two ways to look at the output of the code you have written in the Do file:
 - ① Just execute the codes in the Do file and look at the STATA output window.
 - ② Create a corresponding log file of the Do file.
 - ③ **Always do the second one!!!**
- A log-file keeps a record of all the commands you put into Stata AND the resulting Stata output during the period that the log-file is open.
- Stata will automatically save log-files with the extension **.smcl**, but you can choose an alternative extension, such as **.txt** (recommended for this class).
- Stata will save your log-files in the current working directory.

We can also write .log files into our .do file

```
* My first do-file by Kitak
*make folder "EC2228\week2"
cd "$wd\Week02"
*start log file
log using "wk2_log.log", replace
...
*close log file
log close
```

Data Types

- **Cross Section:** Different units of observation (car models) at a point in time (1978) **webuse auto, clear**
- **Time Series:** Same unit of observation (GDP) at different points in time (1970–2020) **bcuse consump, clear**
- **Panel:** Different units of observation (car models) at different points in time (2000–2020). E.g. a dataset that contains the prices of different cars for each of the years **bcuse cornwell, clear**
- **Why should we pay any attention to data structures?**

Joining datasets

- Often the variables you need live in **different files**.
- A **join** combines two datasets on a common key variable (e.g. **id**).
- In Stata, the command is **merge**. The **keep()** option decides which rows survive:
 - 1 **keep(match)** → **inner join**: only rows present in *both* files.
 - 2 **keep(master match)** → **left join**: all rows from the master file.
 - 3 **keep(using match)** → **right join**: all rows from the using file.
 - 4 **keep(master using match)** → **outer join**: all rows from either file.

Joining datasets: example

Two small files, matched on **id**:

Master (name)

id	name
1	Alice
2	Bob
3	Carol
4	Dave

Using (score)

id	score
1	85
3	92
4	78
5	99

```
use "join_master", clear
merge 1:1 id using "join_using", keep(match) nogen
```

Joining datasets: what each join keeps

- **Inner join** (match): ids 1, 3, 4.
- **Left join** (master match): ids 1, 2, 3, 4; Bob's score is missing (.).
- **Right join** (using match): ids 1, 3, 4, 5; id 5 has no name.
- **Outer join** (master using match): ids 1, 2, 3, 4, 5.

The **nogen** option suppresses the `_merge` variable that Stata would otherwise create to flag where each row came from.

Stata help

- If you ever forget how to use a command... and I often do... type **help** followed by the command.
- For example, type

```
help describe
help log
```

Week 2 Class Assignment

Call the **wage1** dataset from **bcuse**.

- 1 What is the median value of the **educ** variable? What command line would you use?
- 2 Compute the correlation between **educ** and **exper**. What command line would you use?

Generate a log file and upload it on Canvas.